

OFFENSIVE WORDS REFERENCE — Reconciled with Real LIME & SHAP Results

Hate Speech Detection in Text — Research Documentation | Qurat ul Ain Saeed (MSAIF22M017), University of the Punjab, Lahore

CONTENT WARNING: This document lists explicit hate-speech and offensive terms extracted from the RUHSOLD dataset, for academic research, transparency, and reproducibility only.

Purpose. This reference reconciles the original ID↔word catalogue with the **actual** tokens identified by the trained BERT+CNN-Gram model using LIME (num_samples = 1000) and SHAP (word-level masker, 100 samples). It replaces the earlier, illustrative score column with real, computed values and adds the high-impact tokens that the model actually relied on.

1. Verified Model Performance (replaces earlier figures)

Metric	Value	Note
Accuracy / Macro-F1	0.91	was reported 0.89/0.90
ROC-AUC	0.96	was reported 0.93
MCC	0.82	was reported 0.86
Dataset	RUHSOLD, 10,013 posts	46.6% Hate / 53.4% Non-Hate (whole set)
Significance vs best baseline	McNemar p = 0.0015	proposed model significantly better

2. Original ID Catalogue — corrected with real scores where the word appears in model output

ID	Word	Real LIME (summed)	Real SHAP imp.	Context / Usage
ID001	bhenchod	−0.83	0.84	Strong slur (variants: bc, bhenchood)
ID002	gandu	not in top features	—	Offensive slur
ID003	randi	−0.96	0.41	Highly offensive (variant: randiyan, rundi)
ID004	kutta	not in top features	—	Context-dependent (dog)
ID005	pagal	—	—	Context-dependent (crazy/mad); frequent false positive
ID011	saali	—	0.82	Gender-specific, context-sensitive
ID013	lanattt	−1.01 (as laanat)	0.49 (as lanat)	Curse, stylized spelling

All other original IDs (ID006–ID010, ID012, ID014–ID027) retain their original words; they did not appear among the model’s top features in this run and carry no computed score.

3. Real Top Hate-Indicative Tokens (this study) — new IDs ID028+

Tokens the model actually weighted most toward the Hate class but absent from the original 27-word table. LIME = summed weight (more negative = stronger hate signal); SHAP = mean |SHAP| for the Hate class. Rows shaded yellow are identity / named-entity terms — see the bias note.

ID	Word	LIME	SHAP	Type / Note
ID028	bc	−1.46	0.73	Abbreviated slur (behen-chod)

ID029	bharwe	−1.04	0.85	Slur (pimp)
ID030	laanat	−1.01	0.49	Curse
ID031	bhaunk	−0.86	0.43	Slur (bark)
ID032	phattu	−0.70	—	Insult (coward)
ID033	qandel	−0.51	—	Named individual (Qandeel Baloch) — bias signal
ID034	chutiya	−0.48	0.46 (chod)	Slur (fool/idiot)
ID035	ahmaq	−0.40	—	Insult (fool)
ID036	harami	−0.39	—	Slur (bastard)
ID037	marasi	−0.38	—	Caste-based slur
ID038	paidaishi	−0.33	—	Intensifier (“by birth”)
ID039	ahmedi	−0.32	—	Religious minority (Ahmadi) — bias signal
ID040	gashti	−0.31	—	Slur (gender-based)
ID041	randiyan	—	0.87	Slur (plural of randi)
ID042	saalay	—	0.83	Insult
ID043	bhosdiwale	—	0.80	Slur
ID044	champli	—	0.68	Slur
ID045	bhadwaa	—	0.63	Slur (pimp)
ID046	bosdk	—	0.60	Slur (bhosdike, shortened)
ID047	india	—	0.54	National reference — bias signal
ID048	hijra	—	0.49	Transgender community — bias signal
ID049	rundi	—	0.44	Slur (variant of randi)
ID050	addara	—	0.42	Regional slang term

4. Bias Note (identity & named-entity features)

Both explainers surfaced terms that are **not** inherently abusive yet were weighted heavily toward the Hate class: **ahmedi** (Ahmadi religious minority), **hijra** (transgender community), **india** (national reference), and the named individual **qandel** (Qandeel Baloch). This indicates the model can over-rely on the mention of a protected group or person, risking false positives on neutral discussion of those communities. This is documented as a limitation in the thesis (Section 4.6) and motivates fairness-aware training and group-level error auditing.

5. Methodology Note

LIME scores are summed token weights over 50 explained test instances (25 per class) at `num_samples = 1000`; the sign follows the dataset label convention (0 = Hate, so negative = pushes toward Hate). SHAP values are mean $|\text{SHAP}|$ for the Hate class using a word-level text masker over 100 test instances. Source code: [All_Models_Comparison_Colab.ipynb](#) and [hate_speech_fixed_pipeline_v2.py](#).